Homework 7
Numerical Analysis
(MATH 461) Fall, 2023

Due: 11/03/2023

Problem 1. Use the known values of the function $\sin(x)$ at $x = 0, \pi/6, \pi/4, \pi/3, \pi/2$ to derive an interpolating polynomial $p(x)$. What is the degree of your polynomial? What is the interpolation error magnitude $|p(1.2) - \sin(1.2)|$?

Problem 2. Given $n+1$ data pairs $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$, define, for $j = 0, 1, \ldots, n$, the quantities $\rho_j = \prod_{i \neq j} (x_j - x_i)$ and the function $\psi(x) = \prod_{i=0}^{n} (x - x_i)$.

1. Show that $\rho_j = \psi'(x_j)$
2. Show that the interpolating polynomial of degree at most n, denoted p_n, is given by

$$p_n(x) = \psi(x) \sum_{j=0}^{n} \frac{y_j}{(x - x_j)\psi'(x_j)}.$$

Problem 3. A strong benefit of Newton's representation of an interpolating polynomial is its adaptive nature. That is, if one writes the unique polynomial of degree at most n interpolating the $n+1$ points

$$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$$

in Newton's form and desires to add another point (x_{n+1}, y_{n+1}), then it is cheap to construct the unique polynomial of degree at most $n+1$ that interpolates the $n+2$ points

$$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n), (x_{n+1}, y_{n+1}).$$

In practice, one occasionally encounters the reverse situation where one needs to remove a point. That is, suppose $p(x)$ is the unique polynomial of degree at most $n+1$ interpolating the $n+2$ points

$$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n), (x_{n+1}, y_{n+1}),$$

and suppose we want to cheaply construct the unique polynomial of degree at most n, call it $q(x)$, interpolating the $n+1$ points

$$(x_0, y_0), \ldots, (x_{i-1}, y_{i-1}), (x_{i+1}, y_{i+1}), \ldots, (x_n, y_n).$$

If $p(x)$ and $q(x)$ are written in Newton's form

$$p(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1) + \ldots + c_n(x - x_0)\ldots(x - x_n)$$

$$q(x) = k_0 + k_1(x - x_0) + \ldots + k_i(x - x_0)\ldots(x - x_{i-1}) + k_{i+1}(x - x_0)\ldots(x - x_{i+1}) + \ldots + k_{n-1}(x - x_0)\ldots(x - x_{i-1})(x - x_{i+1})\ldots(x - x_n),$$

describe a method to obtain the coefficients $k_0, \ldots, k_{n-1}$ from the coefficients $c_0, \ldots, c_n$ and the points $x_0, \ldots, x_{n+1}$.

Problem 4. Construct two simple examples for any positive integer n, one where interpolation at $n+1$ equidistant points is more accurate than interpolation at $n+1$ Chebyshev points and one where Chebyshev interpolation is more accurate and clearly explain how you know this. Your examples should be convincing without the aid of any computer implementation.